DATASETS

Most popular Datasets

Kaggle datasets

Website: https://www.kaggle.com/

- Kaggle, a subsidiary of Google LLC, is an online community of data scientists and machine learning practitioners. Kaggle allows users to find and publish data sets, explore and build models in a web-based data-science environment, work with other data scientists and machine learning engineers, and enter competitions to solve data science challenges.
- Inside Kaggle you'll find all the code & data you need to do your data science work. Use over 19,000 public datasets and 200,000 public notebooks to conquer any analysis in no time.
- Tour of Kaggle's website website (Datasets, compete, notebooks, discuss, courses, and more)

UCI ML Repository

Website: http://archive.ics.uci.edu/ml/index.php

- University California at Irvine Machine Learning Repository
- It traditionally hosts machine learning datasets used by the machine learning research community.

Criticisms of the Repository

- The datasets are cleaned, meaning that the researchers that prepared them have often already performed some pre-processing in terms of the selection of attributes and instances.
- The datasets are small, this is not helpful if you are interested in investigating larger scale problems and techniques.
- Datasets are limited to tabular data, primarily for classification (although clustering and regression datasets are listed). This is limiting for those interested in natural language, computer vision, recommender and other data.

A self-study plan

For each problem, go through the following steps in the machine learning process:

- Define the problem
- Prepare data
- Evaluate algorithms
- Improve results
- Write-up results

It is a good idea to work on machine learning problems that matter to you.

Data Preparation Process

How to prepare data for a machine learning algorithm. This is a big topic and you will cover the essentials.

The more disciplined you are in your handling of data, the more consistent and better results you are like likely to achieve.

The process for getting data ready for a machine learning algorithm can be summarized in three steps:

- Step 1: Select Data
- Step 2: Preprocess Data
- Step 3: Transform Data

Step 1: Select Data

This step is concerned with selecting the subset of all available data that you will be working with.

Below are two questions to help you think through this process:

- What data don't you need to address the problem?
- What data is not available that you wish you had available? For example data that is not recorded or cannot be recorded. You may be able to derive or simulate this data.

It is only in small problems, like competition datasets where the data has already been selected for you.

Step 2: Preprocess Data

This preprocessing step is about getting the selected data into a form that you can work.

Three common data preprocessing steps are formatting, cleaning and sampling:

- Formatting: The data you have selected may not be in a format that is suitable for you to work with. The data may be in a relational database and you would like it in a flat file, or the data may be in a proprietary file format and you would like it in a relational database or a text file.
- Cleaning: Cleaning data is the removal or fixing of missing data. There may be data instances that are incomplete and do not carry the data you believe you need to address the problem. These instances may need to be removed. Additionally, there may be sensitive information in some of the attributes and these attributes may need to be anonymized or removed from the data entirely.

- Sampling: There may be far more selected data available than you need to work with. More data can result in much longer running times for algorithms and larger computational and memory requirements. You can take a smaller representative sample of the selected data that may be much faster for exploring and prototyping solutions before considering the whole dataset.